

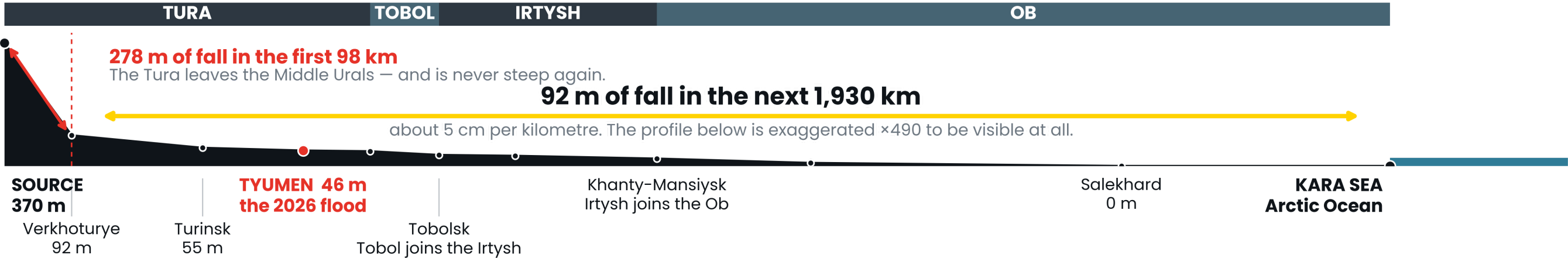
OUT OF DOWNHILL

The Tura rises at 370 m in the Middle Urals and reaches the Kara Sea through the Tobol, the Irtysh and the Ob. **It does four-fifths of its falling in the first 98 km.** The swamps that hold half of Russia's marshland, a 3,650 km river with a single dam, an estuary a thousand kilometres long, a water conflict with China, and the flood that put my own city under in August 2026 all follow from what happens next: a gradient of about five centimetres per kilometre. I measured that gradient from satellite elevation data, then measured what it did to Tyumen.

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Valley-floor elevation from Copernicus DEM GLO-30, measured for this poster · straight-line distance from source

channel lengths: Tura 1,030 km · Tobol 1,591 km · Irtysh 4,248 km · Ob 3,650 km



1 WHAT THE GRADIENT BUILDS

Upper course — 370 m to 92 m

Steep gradient, high potential energy, so erosion is **vertical**: hydraulic action and abrasion cut down faster than the sides retreat, giving a narrow V-shaped valley and interlocking spurs. Load is coarse and angular.

Middle course — the plain

The gradient collapses and **lateral erosion** takes over. The Tura meanders, undercutting outer banks, building point bars on inner ones, widening a floodplain it then abandons in stages. Those abandoned floodplains are the **river terraces** Tyumen stands on — and the reason one bank of the city floods and the other does not.

Lower course — the Ob

Valley 30–50 km wide, floodplain 20–30 km, and below the Irtysh the spring flood spreads **40–50 km** across it: the second largest floodplain on Earth. It ends in a delta of 4,000 km² and the **Gulf of Ob** — at 800–1,000 km the world's longest estuary, and only 10–12 m deep.

Why so much of it is swamp

With almost no slope the channels cannot evacuate meltwater or rain, so it sits on the interfluvies. **Half of all the marshland in Russia is in this basin.** A seventh of the catchment drains internally and never reaches the Ob at all.

The Ob also flows **north**. The south thaws first, so meltwater arrives while the lower river is still sealed under 2 m of ice — it backs up and spreads sideways.

2 A RIVER AT WORK

People

27 million people, 39 cities over 100,000. The fields of the middle and lower Ob give about **two-thirds of Russia's oil and gas**. Navigable ~190 days a year upstream, 150 downstream, 100–110 in the Gulf; warming is adding **5.3 days a decade**.

Energy — the gradient again

3,650 km of Ob carries **one hydroelectric station**: Novosibirsk, 490 MW, whose reservoir floods 1,070 km². Flat ground buys no head and drowns huge areas. The Irtysh, falling out of the Altai, carries three larger ones — Shulbinsk 702 MW, Bukhtarma 675 MW, Ust-Kamenogorsk ~330 MW.

Conflict: whose water is the Irtysh?

It is **the longest transboundary tributary in the world** — rising in China, crossing Kazakhstan, entering Russia. China's **Irtysh-Karamay canal**, begun 1994, already takes over **2.5 km³ a year** and is built for 5–7. Downstream it is drinking water for four million Kazakhs.

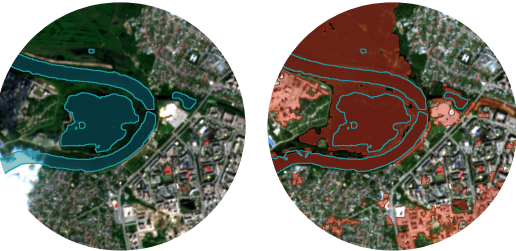
There is still **no allocation agreement**. China has signed neither the UN Watercourses nor the UNECE Water Convention, and under its law Russia is not a party at all — the river never crosses their shared border.

Nature

The **Vasyugan Mire** — 53,000–55,000 km², the largest mire system in the northern hemisphere — sits on the Ob-Irtysh watershed and holds 400 km³ of fresh water. Only ~11% is protected. The Ob estuary feeds up to **3 million migrating birds**. Sturgeon catch fell from **1,410 tonnes in the 1930s to 11 by 1997**, 40% of the spawning grounds cut off by the dams above.

3 WHAT I MEASURED AT TYUMEN

Rain on the headwaters in the Middle Urals. Seven hundred kilometres downstream the Tura rose for three weeks through my home city, peaking at **897 cm on 2 August 2026** — a record summer flood. I filled sandbags for two days, then mapped it from **Sentinel-2**.



1 in 6 buildings flooded on the low north bank, against **1 IN 250** on the high south bank.

18 JUNE
137 ha of water

29 JULY
415 ha — **the river tripled**

TYUMEN, the Tura

old town · the ground we added



SHREWSBURY, the Severn

old town · the ground we added



The 1586 fort took the bluff 36 m up; everything added since sits 11 m up. The same section through **Shrewsbury** puts the old town 27 m up and Frankwell — flooded three times in six weeks in 2000 — 11 m up. **6.3 million English properties are at risk now; 8 million by mid-century.**

Method. Long profile: lowest Copernicus DEM GLO-30 elevation within 1.8 km of each waypoint, measured for this poster; distances are straight-line, so shorter than the channel. Flood extent: Sentinel-2 L2A, 18 June and 29 July 2026, MNDWI (Xu 2006) constrained by elevation and by connectivity to the channel; exposure from OpenStreetMap and WorldPop. GLO-30 is a surface model, so no flood depth is modelled for Shrewsbury, and the 29 July image precedes the peak — Tyumen's figures are a lower bound. **Sources.** Britannica; WWF/TNC Freshwater Ecoregions of the World; Eurasian Development Bank, *The Irtysh River Basin* (2025); UNESCO tentative list 5114; Ramsar SIS; CITES AC16 7.2; Yang, Ye & Shiklomanov, *J. Hydromet.*; Ivanov et al., *Land* (2022); Xu et al., *Env. Res. Comms* (2026); RusHydro; Samruk-Energy; Kommersant; The Moscow Times; Shropshire Council gauge board; Environment Agency NaFRA (2024). **Images** © European Union / ESA (Copernicus Sentinel-2 and DEM GLO-30), free and open licence. Every map, section and figure here is my own work from those data; the code, QGIS project and derived layers are public.